Class Test 1

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Regular Expressions and Validation in JavaScript

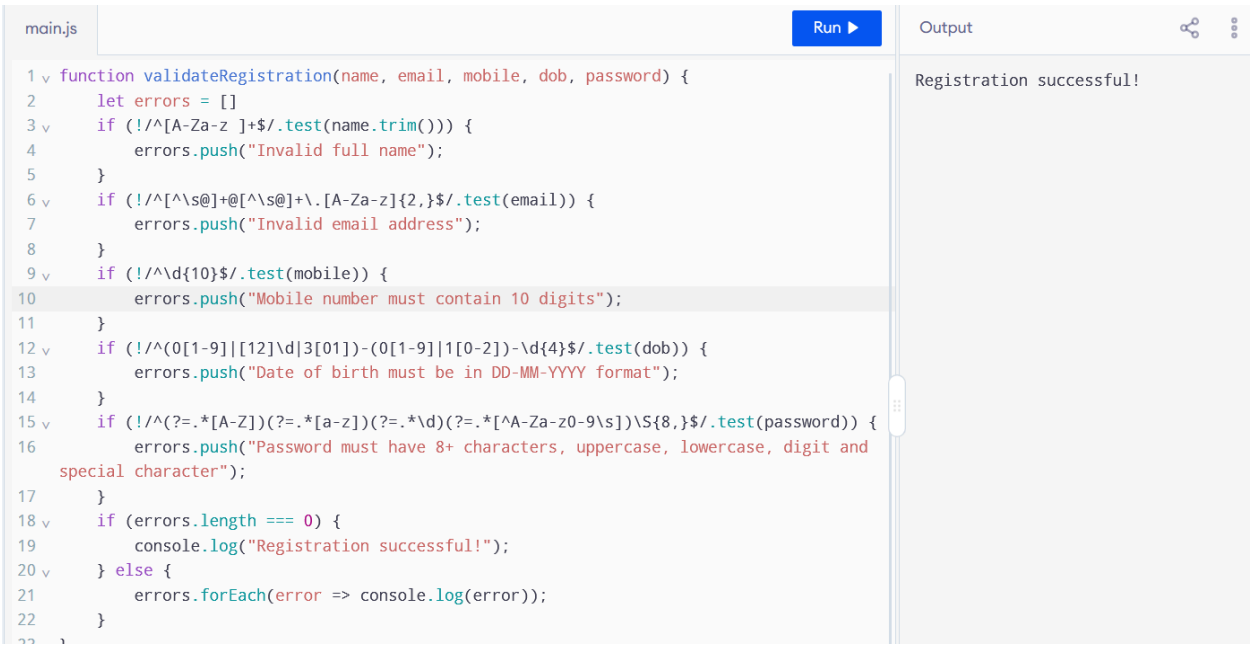
1) Student Registration Validation: Write a JavaScript program to validate a student registration form containing a full name, email address, 10-digit mobile number, date of birth in DD-MM-YYYY format, and password. The password must contain at least one uppercase letter, one lowercase letter, one digit, and one special character, with a minimum length of eight characters.

A.

```
function validateRegistration(name, email, mobile, dob, password) {
  let errors = []
  if (!/^[A-Za-z ]+$/.test(name.trim())) {
    errors.push("Invalid full name");
  }
  if (!/^[^\s@]+@[^\s@]+\.[A-Za-z]{2,}$/.test(email)) {
    errors.push("Invalid email address");
  }
  if (!/^\d{10}$/.test(mobile)) {
    errors.push("Mobile number must contain 10 digits");
  }
  if (!/^(0[1-9]|[12]\d|3[01])-(0[1-9]|1[0-2])-\d{4}$/.test(dob)) {
    errors.push("Date of birth must be in DD-MM-YYYY format");
  }
  if (!/^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[^A-Za-z0-9\s])\S{8,}$/.test(password)) {
    errors.push("Password must have 8+ characters, uppercase, lowercase, digit and special character");
  }
  if (errors.length === 0) {
    console.log("Registration successful!");
```

```
    } else {
        errors.forEach(error => console.log(error));
    }
}
validateRegistration(
    "Arun Kumar",
    "arun@gmail.com",
    "9876543210",
    "15-08-2004",
    "Arun@123"
);
```

OUTPUT:

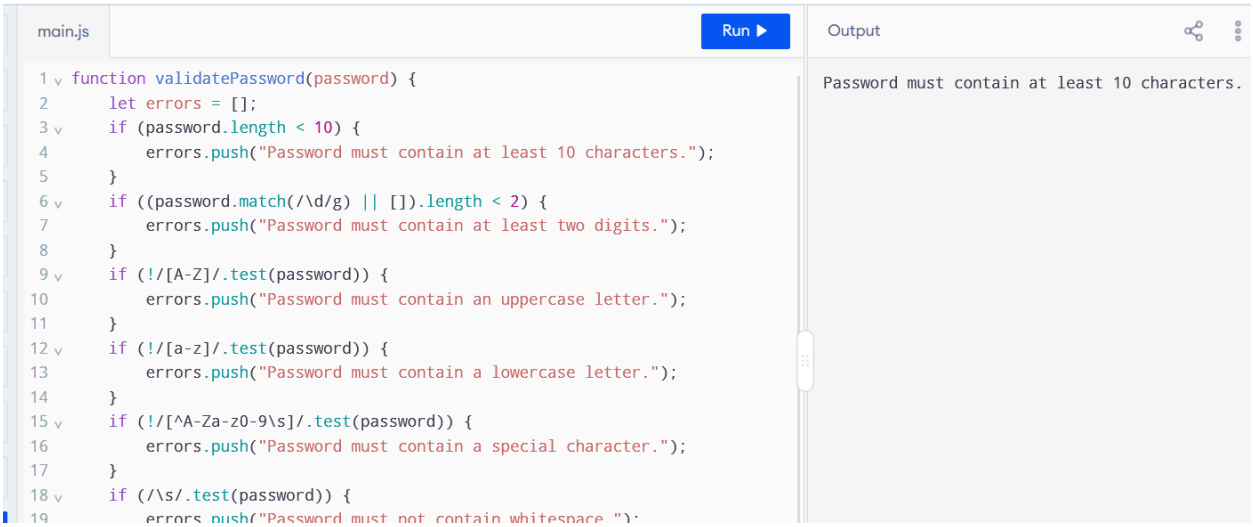


2) Password Strength Validation: Write a JavaScript program to validate a password with the following rules: minimum 10 characters, at least two digits, one uppercase letter, one lowercase letter, one special character, and no whitespace. Display an appropriate error message for each failed condition.

A.

```
function validatePassword(password) {
    let errors = [];
    if (password.length < 10) {
        errors.push("Password must contain at least 10 characters.");
    }
    if ((password.match(/\d/g) || []).length < 2) {
        errors.push("Password must contain at least two digits.");
    }
    if (!/[A-Z]/.test(password)) {
        errors.push("Password must contain an uppercase letter.");
    }
    if (!/[a-z]/.test(password)) {
        errors.push("Password must contain a lowercase letter.");
    }
    if (!/[^A-Za-z0-9\s]/.test(password)) {
        errors.push("Password must contain a special character.");
    }
    if (/\s/.test(password)) {
        errors.push("Password must not contain whitespace.");
    }
    if (errors.length === 0) {
        console.log("Strong password!");
    } else {
        errors.forEach(error => console.log(error));
    }
}
validatePassword("Abc@12345");
```

OUTPUT:

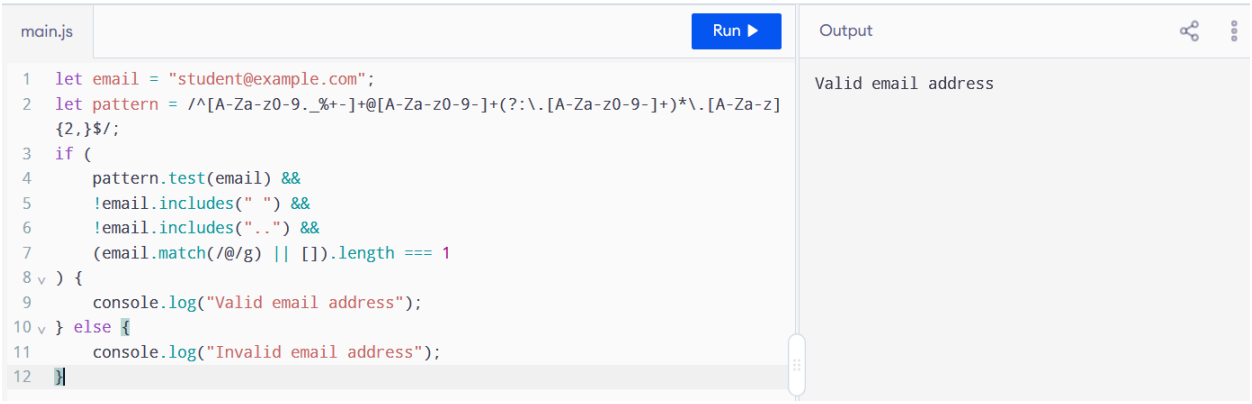


3) Email Address Validation: Write a JavaScript program to validate email addresses according to these rules: the address must contain exactly one @, a non-empty username, a domain name, and a valid-looking extension of at least two letters. Reject spaces and consecutive dots.

A.

```
let email = "student@example.com";
let pattern = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9-]+(?:\.[A-Za-z0-9-]+)*\.[A-Za-z]{2,}$/;
if (
    pattern.test(email) &&
    !email.includes(" ") &&
    !email.includes("..") &&
    (email.match(/@/g) || []).length === 1
) {
    console.log("Valid email address");
} else {
    console.log("Invalid email address");
}
```

OUTPUT:

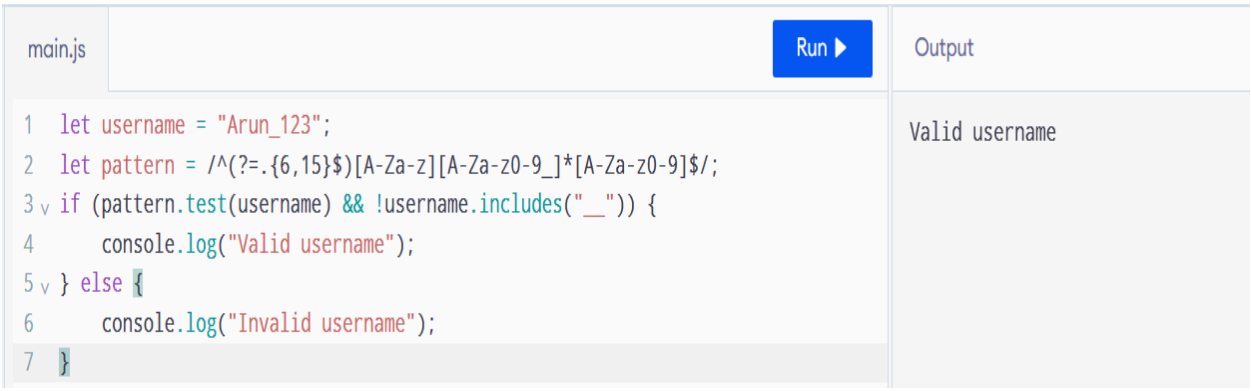


4) Username Validation: Write a JavaScript program to validate a username that must start with a letter, contain only letters, digits, and underscores, have a length of 6–15 characters, and must not end with an underscore or contain consecutive underscores.

A.

```
let username = "Arun_123";
let pattern = /^(?=.{6,15})[A − Za − z][A − Za − z0 − 9_] * [A − Za − z0 − 9]/;
if (pattern.test(username) && !username.includes("__")) {
    console.log("Valid username");
} else {
    console.log("Invalid username");
}
```

OUTPUT:



5) Extract and Analyse Data: Write a JavaScript program to extract all mobile numbers, email addresses, and dates from a paragraph using regular expressions. Display each category separately.

A.

```
let paragraph = `
Contact Arun at 9876543210 or arun@gmail.com.
Meet Priya on 25-12-2025.
Her mobile number is 9123456780 and email is priya@yahoo.in.
The next meeting is on 05-01-2026.
`;
let mobiles = paragraph.match(/\b[6-9]\d{9}\b/g) || [];
let emails = paragraph.match(
  /\b[A-Za-z0-9._%+-]+@[A-Za-z0-9-]+(?:\.[A-Za-z0-9-]+)*\.[A-Za-z]{2,}\b/g
) || [];
let dates = paragraph.match(
  /\b(?:0[1-9]|[12]\d|3[01])-(?:0[1-9]|1[0-2])-\d{4}\b/g
```

```
) || [];
console.log("Mobile Numbers:");
console.log(mobiles);
console.log("Email Addresses:");
console.log(emails);
console.log("Dates:");
console.log(dates);
```

OUTPUT:

